MIL-F-80207B
20 August 1984
SUPERSEDING
MIL-F-80207A
28 March 1978

MILITARY SPECIFICATION

FURNACES, HEAT TREATING, HORIZONTAL, CAR-BOTTOM DESIGN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers horizontal, car-bottom heat treating furnaces of the forced convection and direct heated design.

1.2 Classification.

1.2.1 Types, styles, and classes. The furnaces shall be of the following types, styles, and classes as specified (see 6.2.1).

Type I - Electrically Heated

Type II - Gas Fired

Style A - Forced Convection
Class 1 - 300° - 1000°F Operating Temperature
Class 2 - 300° - 1250°F Operating Temperature
Class 3 - 300° - 1400°F Operating Temperature

Style B - Direct Heated
Class 1 - 400° - 1800°F Operating Temperature
Class 2 - 1200° - 2000°F Operating Temperature
Class 3 - 1200° - 2400°F Operating Temperature

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-TES, Memphis, Tennessee 38114-5297 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.2 Size. The size of the furnace chamber shall be as specified (see 6.2.1). The furnace chamber size shall start from a minimum of three feet in width by four feet in length by three feet in height. Size changes shall increase in increments of six inches in width or height and 12 inches in length, as required. The dimensions specified shall be the minimum inside effective working area of the heating chamber in which temperature uniformity and performance requirements specified shall be maintained. The chamber shall be capable of accommodating a charge equal in size to not less than the dimensions specified.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-3296 - Forges, Furnaces, and Ovens (Exclusive of Space Heating and Cooking, Packaging of)

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI C96.1 - Temperature Measurement Thermocouples

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

JOINT INDUSTRIAL COUNCIL

JIC EGP-1 - Electrical Standards for General Purpose Machine Tools

(Applications for copies should be addressed to the Joint Industrial Council, 7901 Westpark Drive, McLean, VA 22101.)

FACTORY MUTUAL ENGINEERING CORPORATION (FM)

Handbook of Industrial Loss Prevention

(Application for copies should be addressed to Factory Mutual Engineering Corporation, 1151 Boston-Providence Turnpike, Norwood, MA 02062.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-3951 - Commercial Packaging, Standard Practice for

ASTM-E-380 - Metric Practice Guide

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION

AGMA 360.02 - Manual For Machine Tool Gearings

AGMA 390.03 - Gear Classification, Materials and Measuring Methods For Unassembled Gears

(Application for copies should be addressed to the American Gear Manufacturers Association, Standards Department, Suite 1000, 1901 North Fort Myer Drive, Arlington, VA 22209-1695).

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

ISO - 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametral Pitches of

ISO - R1122 - Gears, Glossary of, Geometrical Definitions

(Applications for copies should be addressed to the International Organization for Standardization, c/o American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Design. The furnaces shall be new and of the manufacturer's current design. The furnace shall be of the horizontal car bottom design with the hearth and bottom of the furnace mounted on wheels for moving outside the furnace chamber for loading and unloading. The furnace shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. Unless otherwise specified (see 6.2.1), furnaces with working widths up to five feet shall be completely shop erected and lined in the vendors plant and furnaces with a working width in excess of five feet shall be field erected at the installation specified.

3.1.1 Type I. The type I furnace shall be electrically heated and operate from a single service power source. Unless otherwise specified (see 6.2.1), the furnace shall operate on a primary power supply of 440 volts, 60 Hz, 3 phase power supply.

3.1.2 Type II. The type II furnace shall be gas fired and, unless otherwise specified (see 6.2.1), designed to operate on a fuel supply of 1000 BTU/Cubic Foot of natural gas.

3.1.3 Style A. The style A furnace shall employ the forced convection method of heating. Excess air burners or circulating fans shall be provided as necessary to meet the temperature uniformity and heating rates specified for the furnaces. Unless otherwise specified, the product of combustion may mix with circulating air in the work chamber. When required (see 6.2.1), the air shall be heated in an area separated from the charge and blown into the work chamber by one or more circulating fans. The charge shall not be exposed to the products of combustion or exposed to the radiant heat emitted by the heat source.

3.1.4 Style B. The style B furnace shall be of the direct heated design with the work directly exposed to the heat source. Excess air burners, high velocity burners or circulating fans shall be provided as necessary to meet the temperature uniformity and heating rates specified for the furnace.

3.1.5 Safety and health requirements. Each furnace shall be provided with safety features and devices for the protection of personnel, equipment and the work being processed. In addition to other safety requirements specified herein, the safety system of the furnace shall conform to the applicable requirements of FM "Handbook of Industrial Loss Prevention". All

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furnace parts, components, mechanisms, and assemblies furnished on the furnace, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the furnace itself. Additional safety and health requirements shall be as specified (see 6.2.1).

3.1.6 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the furnace. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in ASTM-E-380.

3.1.7 Reclaimed materials. The furnace may contain reclaimed materials provided such materials will not jeopardize the furnace's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the furnace.

3.1.8 Energy efficiency. The furnace and all its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.1.9 Environmental protection. The furnace shall be designed and constructed so that, under the operating, service, transportation and storage conditions described herein, the furnace shall not emit materials hazardous to the ecological system as prescribed by Federal, state or local statutes in effect at point of installation.

3.1.10 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings and filler caps shall be accessible.

3.1.11 Interchangeability. To provide for replacement of worn or damaged parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or furnace.

3.2 Construction. The furnace shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the furnace to its maximum rating and capacity without permanent distortion.

3.2.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding,

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peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the furnace. Such processes may be used only for enhancing surface finish and appearance.

3.2.2 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.2.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed.

3.2.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.2.5 Painting. All surfaces to be painted shall be cleaned of all foreign matter. Unless otherwise specified (see 6.2.1), the furnace shall be painted in accordance with manufacturer's standard commercial practice which shall be not less than one coat of primer and one coat of finish color.

3.2.6 Threads. All threaded parts used on the furnace and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.2.7 Gears. All gears and pinions used in the furnace or its components, shall be properly selected for the intended application and fabricated to either English (USA) standard inch diametral pitch per AGMA 390.03, AGMA 360.02 or Metric (SI) Module ratio per the International Organization for Standardization, ISO-R1122, ISO-54, and ASTM-E-380.

3.3 Electrical equipment. Except as otherwise specified herein, electrical equipment and installation shall conform to JIC Standard EGP-1.

3.3.1 Motors. All motors shall be totally enclosed or located in totally enclosed areas. Motors rated in excess of 1/2 horsepower shall be dual voltage, 230/460 volt, 60 Hz, 3 phase type.

3.3.2 Electrical connections. Electrical connections of conductors and terminal parts shall be of the pressure or solder type. When soldered connections are used, the conductors and terminal parts shall be mechanically

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secured before soldering. Flux for soldering shall be of a resin base.

3.4 Performance. The furnace shall be capable of operating continuously at any desired temperature between the lowest and highest temperature specified for the class furnace.

3.4.1 Nature of material. The nature of the material that the furnace shall be designed to heat, such as steel, brass, or aluminum, shall be specified (see 6.2.1).

3.4.2 Type of process. The type of process such as hardening, drawing, or tempering, and the minimum and maximum dimensions of the material to be processed shall be specified (see 6.2.1).

3.4.3 Load capacity. The load capacity of the furnace shall be specified (see 6.2.1), and shall be based on pounds per square foot of hearth area. Hearth area is defined as the width and length dimension specified for the size furnace.

3.4.4 Heating capacity. The furnace shall be capable of heating a capacity charge from 77°F, ±18°F, to maximum operating temperature in not greater than the time limit specified (see 6.2.1). Heating capacity shall be based on the furnace soaked out at 300°F when the charge enters the chamber.

3.4.5 Temperature uniformity. With the furnace charged and the chamber temperature stabilized, the temperature variation from control point to any other point within the work zone shall not vary more than plus or minus the number of degrees Fahrenheit specified (see 6.2.1). The furnace shall be capable of maintaining the temperature uniformity specified with any size charge up to maximum load and at any desired temperature control point within the operating temperature range specified for the class furnace.

3.5 Components and systems.

3.5.1 Shell. The outer shell of the style A, class 1 furnace shall be fabricated from steel plate not less than 0.074 inch thick (Manufacturer's Standard Gage No. 14). The shell of the style A, class 2 and 3 furnaces shall be fabricated from steel plate not less than 0.119 inch thick (Manufacturers Standard Gage No.11). The shell of all style B furnaces shall be fabricated from steel plate not less than 0.179 inch thick (Manufacturer's Standard Gage No. 7). The shell shall be reinforced with buck stays and cross tied with "I" beams and angle shapes to provide a rigid structure. Welded construction shall be used throughout, except any necessary field erection joints which shall be bolted. Design and construction shall provide for the protection of all furnace components exposed to chamber temperature. When necessary, the front plate shall be water cooled in all areas that will be exposed to the radiant heat when the door is in the raised position.

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3.5.2 Insulation.

3.5.2.1 Style A, class 1. Unless otherwise specified (see 6.2.1), the style A, class 1 furnace shall be lined with aluminized steel plate or type 304, 18-8 stainless steel not less than 0.074 inch thick (14 gage) and insulated with fireproof industrial rock wool, ceramic fiber or similar non-powdering, non-setting insulating material. The overall quality and thickness of the insulation shall be such that the outside surface temperature of the chamber will not exceed 80°F above ambient air temperature with the chamber temperature stabilized at maximum temperature.

3.5.2.2 Style A, class 2 and class 3 and all style B. The lining of style A class 2 and class 3 furnaces shall consist of not less than 4 inches of 8 lbs/cu ft density, 2300°F ceramic fiber, backed up with not less than 4 inches of 4 lbs /cu ft density, 1700°F ceramic fiber. The lining of all style B furnaces shall consist of not less than 4 inches of 8 lbs/cu ft density 2600°F ceramic fiber, backed up with not less than 5 inches of not less than 6 lbs/cu ft density, 1700°F ceramic fiber. The lining may consist of either blanket ceramic fiber material or modular ceramic fiber material provided the lining complies with all requirements specified herein. The quality of the ceramic fiber material and the insulation techniques shall be such that the lining will withstand the air velocities, atmospheres and temperatures encountered under all furnace operating conditions without damage to the lining. All lining joints shall be staggered from other joints at least 4 inches. The lining shall be preheated to minimize shrinkage, or compressed to the density recommended by the fiber material manufacturer, when installed to compensate for shrinkage. All installation hardware and installation techniques shall conform with the recommendations of the fiber material manufacturer. With the chamber temperature stabilized at maximum temperature, the outside surface temperature of the style A furnaces shall not exceed 80°F above ambient temperature or the style B furnaces 100°F above ambient temperature. Ambient air temperature is defined as 77°F$\pm$ 18°F in still air.

3.5.3 Hearth and car. The bottom and hearth of the furnace shall form a car mounted on iron wheels with anti-friction bearings. The car shall be self-guided on rails and shall move freely when empty or loaded, hot or cold. The car deck shall be fabricated from steel plate, braced and reinforced with structural members and shapes. The car deck shall be faced with refractory material of a quality and thickness to support capacity loads and protect the car structure from the chamber temperature. The hearth shall be designed with piers, perforated plates, or other devices, as necessary, to provide load distribution and heat circulation to assure the temperature uniformity and heating requirements specified for the furnace. The working height of the car or distance from the top of the hearth to the top of the rail shall not exceed 42 inches. The car shall support capacity loads without any permanent deformation under continuous heating and cooling and loading and unloading. The car bottom mechanism shall be designed so that the car, when traveling into the chamber, will automatically come to a stop against a rigid abutment to assure alignment for sealing and for protection of the furnace.

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3.5.3.1 Car seal. The furnace shall provide a full perimeter seal between the car bottom and furnace walls. Unless otherwise specified, the seal shall be of the ceramic fiber type and sealing shall be automatic when the car is positioned in the chamber. When required (see 6.2.1), seals shall be of the water cooled type or of the sand seal type provided with hydraulic or pneumatic cylinders to raise the seals to the sealing position after the car is positioned in the chamber, as specified. The seals shall minimize heat loss and assure the temperature uniformity specified for the furnace.

3.5.3.2 Car drive system. When specified (see 6.2.1), the car shall be self-propelled and provided complete with direct mounted motor drive system and soft start controls for moving the car in and out of the chamber.

3.5.3.3 Tracks. Unless otherwise specified (see 6.2.1), tracks for furnaces erected in the vendor's plant shall not require recessing in the floor, and tracks for furnaces that are field erected shall not protrude above the floor. The track shall be of sufficient length to provide one full car length plus 4 feet outside the chamber. Additional track length and arrangement shall be provided when specified (see 6.2.1).

3.5.4 Door. Unless otherwise specified (see 6.2.1), furnaces erected in the vendor's plant shall be provided with a manually operated vertical lift type door, and furnaces that are field erected shall be provided with a power operated vertical lift type door. The manually operated door shall be provided with counterweights, and each door shall be mechanically guided. When closed, the door shall seal uniformly to the face place and hearth seal. The door shall be lined with the same type material as the chamber and shall maintain the same outside surface temperature as the furnace walls. With the door in the open position, the clear chamber entrance dimension above the hearth level shall be not less than the width and height specified for the size furnace.

3.5.5 Controls. Controls for the powered door and car puller shall be located in an area shielded from the heat of the open door. The car controls shall be interlocked with the door so that it is impossible to move the car unless the furnace door is completely open. When specified (see 6.2.2), the control shall be provided in a shielded enclosure.

3.5.6 Air circulating system. The furnace shall be designed and constructed with the necessary air ducts, baffles, adjustable vanes, radiation shields, vents, circulating fans, and controls to provide the temperature uniformity and heating rate specified for the furnace. Each fan shall be interlocked with the heating system to prevent operation of the furnace in the event of fan failure. Adequate means shall be provided to assure proper cooling of all bearings and components.

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3.5.7 Heating system.

3.5.7.1 Type I furnaces. The heating elements shall be designed and located to meet the heating requirements and temperature uniformity specified for the furnace. The elements shall be mounted so they will not become loose or work out of place under any operating conditions. Elements for style A, class 1, 2, and 3 furnaces shall be fabricated from an alloy containing approximately 35 percent nickel and 20 percent chromium or from material with as good or better characteristics. Elements for style B, class 1 and 2 furnaces shall be fabricated from an alloy containing 80 percent nickel and 20 percent chromium. Elements for style B, class 3 furnace shall be fabricated from silicon carbide or equivalent material. Furnaces with silicon carbide elements shall be provided with a multiple tap transformer for changing voltage to compensate for change in element resistance.

3.5.7.2 Type II furnaces. The type II furnaces shall be provided with all necessary burners, gas-air mixing systems, blowers, pilots, gauges, and safety devices required for safe continuous operation. Burners shall be sufficient in number and capacity to meet the heating requirements and temperature uniformity specified for the furnace. The burners of direct fired furnaces shall be arranged or located in such a manner that the flame will not impinge on the charge.

3.5.8 Thermocouples. Furnaces having a temperature rating in excess of 2000°F shall be provided with platinum-rhodium type thermocouples and furnaces with a temperature rating of 2000°F and lower shall be provided with chromel-alumel type thermocouples. The thermocouples shall conform to ANSI C96.1. Thermocouple lead wires for furnaces erected in the vendor's plant shall be not less than 50 feet in length. When specified (see 6.2.1), in addition to the thermocouple ports and protective tubes required for furnace control and operation, one additional thermocouple port and protective tube shall be provided within six inches of the control thermocouple port in each control zone.

3.5.9 Multi-zone control. When necessary to maintain the temperature uniformity and performance requirements specified for the furnace, the chamber shall be divided into a suitable number of control zones with the required instruments and controls conforming to 3.5.10 provided for multi-zone control.

3.5.10 Instruments. Unless otherwise specified (see 6.2.1), the furnace shall be provided with an indicating-recording-controlling instrument with a calibrated accuracy of 0.25 percent of full scale reading. Operating characteristics and dimensions shall conform to 3.5.10.1 through 3.5.10.2.1. The temperature controller shall provide thermocouple upscale fail-safe protection that will shut off the heat source when the furnace temperature exceeds approximately 100°F above maximum. The instrument shall be marked to indicate the particular type of thermocouple for which it is calibrated. Automatic reference junction compensation shall be provided in all instruments using thermocouples as the primary temperature sensing element. External thermocouple lead lengths shall not effect calibration of

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the instruments other than for minor adjustments. Indicating instruments shall be designed for mounting on the panels so that any tendencies toward parallax in reading will be alleviated.

3.5.10.1 Temperature controller. Unless otherwise specified (see 6.2.1), the temperature controller for the type I, electrically heated furnace shall be of the current adjusting type and the controller for the type II, fuel fired furnace shall be of the position proportioning control system. The controller for each furnace shall be capable of maintaining the set control temperature within ±2°F within the operating temperature range of the furnace.

3.5.10.1.1 Control indicator scale. The temperature control indicator scale shall be approximately the same length as the temperature recorder chart width. When specified (see 6.2.1), the indicator shall have both fahrenheit and centigrade temperature scales. The indicator shall have a maximum range of not less than 200°F above the maximum temperature range of the furnace. The increments between successive temperature marks shall be not greater than 10°F for furnaces having a range of 1200°F, and not more than 20°F for furnaces having a range in excess of 1200°F. The spacing of numbered marks shall be not greater than 200°F on instruments having a range of 1200°F, and not greater than 400°F on instruments of greater range. The control shall be provided with two pointers. One pointer shall be adjustable to the desired control temperature and the other shall indicate the actual furnace temperature.

3.5.10.2. Temperature recorder. Unless otherwise specified (see 6.2.1), the temperature recorder shall be of the strip chart, self-balancing type.

3.5.10.2.1 Chart. The calibrated chart width shall be not less than 9 inches, and the length shall be not less than 90 feet to the roll. The chart speed shall be two inches per hour and provisions for changing the speed to four and six inches per hour shall be provided. The pen mechanism shall drive the pen across the entire calibrated chart width in not more than 30 seconds. A platen located immediately behind the chart plane shall be furnished to permit the marking of notations on the chart while the instrument is in operation. The chart mechanism shall be furnished in a dust proof housing with a hinged door. The door shall be provided with a glass that will permit full view of the chart width and not less than 10 inches of chart length.

3.5.10.3 Program control. When required, a temperature programmer shall be furnished and integrated with the furnace control system to provide temperature control in accordance with a preselected program as specified (see 6.2.1).

3.5.10.4 Excess temperature controller. When specified (see 6.2.1), the furnace shall be provided with an excess temperature controller. The excess temperature control circuit shall be independent of the control circuitry. The controller shall be of the indicating type with a pointer which may be set at the over-temperature control point and shall function to shut off the heat source to the furnace when the over-temperature control is reached. The excess temperature controller shall provide for furnace shut-down in the

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event of thermocouple failure, or failure of other components in the circuit. In case the furnace is shut down by the excess temperature control system, it shall sound an audible alarm and the furnace shall remain off until started manually. Calibrated accuracy shall be within one percent of full scale reading.

3.5.10.5 Instrument panel. Unless otherwise specified (see 6.2.1), one floor mounted instrument panel conforming to JIC EGP-1 shall be provided with the furnace for mounting all instruments and accessory controls. The accessory controls shall be mounted in a front opening NEMA Type 12 enclosure with a disconnect switch. An enclosure shall be provided for mounting the power contractors separate from the instrument panel. The panel shall provide for semi-flush mounting of all instruments. When specified (see 6.2.1), the instrument panels shall be provided with shock absorbing mounts. Single lights shall be provided, to indicate the furnace cycle performance.

3.5.10.6 Thermocouple burn-out alarm. The excess temperature controller, when furnished, and the temperature controller shall be designed so that loss of either or both thermocoupler circuits will provide an audible and visible alarm.

3.6 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.7 Marking of instruction plates, indicating plates and name plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

3.8 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion resistant metal nameplate, shall be securely attached to each furnace. The nameplate shall contain the information listed below. If the furnace is a special model, the model designation shall include the model of the basic standard furnace and a suffix keyed to the manufacturer's permanent records.

Nomenclature
Manufacturer's name
Manufacturer's serial number
Power input characteristic and ratings
Work zone, dimensions
Load capacity, pounds/sq ft of hearth area
Operating temperature range and uniformity (±°F)
Date of manufacture
Contract or order number
National Stock Number or Plant Equipment Code

3.9 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract.

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3.10 Workmanship. Workmanship of the furnace and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection and tests. The inspection and test requirements are classified as follows:

a. Quality conformance inspection (see 4.3).
b. Acceptance test (see 4.4.).

4.3 Quality conformance inspection. Quality conformance inspection shall be performed on the furnace prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.5, the test in 4.6.1 through 4.6.4, and the inspection in 4.7. Failure of the furnace to pass the examination or any test shall be cause for rejection.

4.4 Acceptance test. Acceptance test shall be performed on the furnace to insure conformance with this specification. Unless otherwise specified (see 6.2.1), the acceptance test shall be performed only after the furnace is installed at its final location, and the acceptance test shall consist of the examination in 4.5 and all tests in 4.6. Failure of the furnace to pass one or more tests or any part of a test will be cause for rejection.

4.5 Examination. The furnace and equipment shall be examined for design, dimensions, construction, materials, components, electrical equipment, safety compliance, and workmanship to determine compliance with the requirements of this specification.

4.6 Tests. All tests shall be conducted at an ambient temperature of 77°F± 18°F. Test instruments and sensing elements used to perform the test specified for the furnace and instruments shall have been checked against a standard potentiometer type instrument and calibrated thermocouple of known accuracy within the previous three months to read accurate within ±2°F.

4.6.1 Instrument test. Indicating and recording instruments shall be tested by impressing various predetermined voltages, simulating thermocouple outputs for different furnace temperatures to terminals where normally the

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thermocouple would be connected. The instruments shall be tested through their operating range. Each control circuit shall be tested to determine compliance with 3.5.10.

4.6.2 Circuit test. Each circuit of the electrical system shall be tested for dielectric strength, continuity, insulation-resistance and faulty grounds in accordance with the requirements of JIC EGP-1.

4.6.3 Performance test. The furnace shall be prepared for operation and operated in accordance with the manufacturer's instructions until the lining and hearth have thoroughly dried. Performance and alignment of all components and systems shall be checked at various temperatures and any necessary adjustments shall be made for correct operation. With the furnace chamber temperature stabilized at 300°F, the car shall be loaded with a capacity charge. The nature of the material shall be as specified for the furnace or have approximately the same heat characteristics and configuration. The gross load shall be equivalent to not less than the weight specified for each square foot of hearth area. The furnace shall heat the charge to maximum operating temperature within the time limit specified for the furnace. With the furnace temperature stabilized at maximum, the temperature of the exterior walls and door shall be measured and shall not exceed the limits specified for the class furnace, except within three inches of points where structural metal extends through the insulation. Each safety device or system shall be tested by simulating a failure that normally would cause the device to operate. Each system shall perform its intended function.

4.6.4 Temperature uniformity test. The furnace shall be loaded with a production type charge equivalent to not less than 80 percent of maximum load, and a chamber temperature survey performed to determine conformance with the uniformity requirements specified for the furnace. A minimum of nine test locations shall be used, with one thermocouple located in each corner of the chamber and one thermocouple located within six inches of the control thermocouple; or, in case of multiple zone control, in addition to the corner thermocouples, one thermocouple shall be located within six inches of each control thermocouple. The survey shall be taken with equilibrium established at minimum and maximum operating temperatures specified for the class furnace. At least three survey readings shall be taken at intervals of five to ten minutes at both minimum and maximum temperature and the temperature variation between the furnace instrument control thermocouple(s) in the chamber and all points under test shall not exceed plus or minus the number of degrees F specified for the furnace.

4.7 Inspection of packaging. Packaging shall be inspected to determine compliance to the requirements of Section 5.

5. PACKAGING

5.1 Packaging, packing and marking. Unless otherwise specified, packaging, packing, and marking shall be in accordance with ASTM-D-3951. When specified, preservation, packing and marking shall be in accordance with

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specification MIL-F-3296. Level of preservation and packing shall be as specified (see 6.2.1).

6. NOTES

6.1 Intended use. Car bottom furnaces are used primarily for heating heavy and bulky material in such processes as annealing, normalizing, forging, and stress relieving.

6.2 Ordering data.

6.2.1 Acquisition requirements. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number, and date of this specification.
b. Type, style, and class furnace required (see 1.2.1).
c. Size furnace required (width, length, height) (see 1.2.2).
d. Erection and installation, if different, and field erection requirements such as location, utilities available, and foundation requirements, when applicable (see 3.1).
e. Power supply, if different (type I only) (see 3.1.1).
f. Fuel supply, if different (type II only) (see 3.1.2).
g. Heating system separate from the recirculating air or gas, when required (see 3.1.3).
h. Specify additional safety and health requirements (see 3.1.5 and 6.4).
i. Measurement system if different (see 3.1.6).
j. Painting, if different (see 3.2.5).
k. Nature of material to be heated (see 3.4.1).
l. Type of process and dimensions of material (see 3.4.2).
m. Load capacity, pounds per square foot of hearth area (see 3.4.3).
n. Heating capacity, maximum time for the furnace to heat a capacity charge (see 3.4.4).
o. Temperature uniformity requirements (degrees F) (see 3.4.5).
p. Lining and insulation, if different, style A, class 1 only (see 3.5.2.1).
q. Power operated sand seals, or water cooled seals, when required (see 3.5.3.1).
r. Car puller, when required (see 3.5.3.2).
s. Track design, if different (see 3.5.3.3).
t. Extra track and arrangement, when required (see 3.5.3.3).
u. Door, if different (see 3.5.4).
v. Shielded enclosure for controls, when required (see 3.5.5).
w. Extra thermocouple port and tube, when required (see 3.5.8).
x. Instruments, if different (see 3.5.10).
y. Temperature controller, if different (see 3.5.10.1).
z. Both fahrenheit and centigrade scales, if required (see 3.5.10.1.1).
aa. Temperature recorder, if different (see 3.5.10.2).
bb. Program control and program performance, if required (see 3.5.10.3).
cc. Excess temperature controller, if required (see 3.5.10.4).

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dd. Instrument panel, if different (see 3.5.10.5).
ee. Shock-absorbing mounting of instrument panel, if required (see 3.5.10.5).
ff. Repair parts, if required (see 3.6).
gg. Nameplate, if different (see 3.8).
hh. Quality conformance inspection, if different (see 4.3).
ii. Acceptance test, if different (see 4.4).
jj. Level of preservation and packing required (see 5.1).

6.2.2 Contract data requirements. All data requirements such as parts lists, wiring diagrams, operating manuals, maintenance manuals, alignment and test results shall be requested on a DD Form 1423, Contract Data Requirements List, and shall be furnished in accordance with each specified DD Form 1664, Data Item Description, as shown on the DD Form 1423.

6.3 Safety and health requirements. Paragraph 3.1.5 requires compliance only with those OSHA requirements that concern the furnace itself. It does not require compliance with those OSHA requirements that concern "the furnace in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, heat, etc. Since OSHA limits the total hazards level of these hazards in the environment (and does not limit the hazard level of individual furnace in the environment) the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional furnace requirements that will integrate the new furnace into its future operating environment. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1.h.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Preparing Activity:
DLA-IP

Project Number:
3424-0103

Review Activity:
Army - AL
Navy - SH
Air Force - 84
DLA - GS

User Activities:
Army - SM, ME
Navy - MC, AS, OS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-80207B	Furnaces, Heat Treating, Horizontal, Car-Bottom Design

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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